

PAPER_837: F_quark + F_neutrino + F_ALP + F_dark — arXiv Bridge to UQFF

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Abstract

Four new $F_{U_{Bi}}$ terms are derived from arXiv high-energy physics papers and integrated into the UQFF Master Buoyancy Equation. F_{quark} (1.54×10^7 N) from Belle II $|V_{cb}|$ measurements is the largest new term. F_{neutrino} (1 N) from neutrino polarizability data, F_{ALP} (10^4 N) from ALP-hadron covariant frameworks, and F_{dark} (1 N) from FASER dark photon searches complete the new physics bridge. All four terms are negligible compared to F_{LENR} (10^{36} - 10^{39} N) in integrated $F_{U_{Bi}}$ calculations, but they represent critical theoretical bridges between Standard Model particle physics and UQFF's beyond-SM framework.

1. New Terms — Origins and Derivations

1.1 F_{quark} — Quark CKM Matrix Coupling (arXiv:2506.15256)

Source: Belle II experiment at KEK, $|V_{cb}| = (39.2 \pm 0.7) \times 10^{-3}$

Physics: CKM matrix element coupling $b \rightarrow c$ quark transition

$$F_{\text{quark}} = k_{\text{quark}} * |V_{cb}|^2$$

$$k_{\text{quark}} = 10^{10} \text{ N} \quad (\text{UQFF coupling constant for quark sector})$$

$$|V_{cb}| = 39.2 \times 10^{-3} = 0.0392$$

$$\begin{aligned} F_{\text{quark}} &= 10^{10} * (0.0392)^2 \\ &= 10^{10} * 1.54 * 10^{-3} \\ &= 1.54 * 10^7 \text{ N} \end{aligned}$$

Physical significance: Represents the CKM-mediated quark flavor coupling to vacuum energy. The $b \rightarrow c$ transition at 1.54×10^7 N is the largest arXiv-derived correction term, reflecting heavy-flavor vacuum polarization.

1.2 F_{neutrino} — Neutrino Polarizability Coupling (arXiv:2506.14881)

Source: NOMAD, MiniBooNE, SBND, DUNE experiments; neutrino polarizability α_ν

Physics: Neutrino electromagnetic polarizability mediating vacuum coupling

$$F_{\text{neutrino}} = k_{\text{neutrino}} * \alpha_\nu$$

$$k_{\text{neutrino}} = 10^{10} \text{ N}$$

$$\alpha_\nu = 10^{-10} \quad (\text{neutrino polarizability, NOMAD-era bound})$$

$$F_{\text{neutrino}} = 10^{10} * 10^{-10} = 1 \text{ N}$$

Physical significance: Neutrino electromagnetic polarizability couples neutrino sea to vacuum energy. Value 1 N is consistent with SM neutrino mass scale $\sim \text{eV}$.

1.3 F_ALP — Axion-Like Particle Hadronic Coupling (arXiv:2506.15637)

Source: ALP-hadron covariant interaction framework

Physics: Axion-like particle coupling to quarks/nucleons via g_{aqq}

$$F_{\text{ALP}} = k_{\text{ALP}} * g_{\text{aqq}}$$

$$k_{\text{ALP}} = 10^{10} \text{ N}$$

$$g_{\text{aqq}} = 10^{-6} \text{ (ALP-quark-quark coupling, below current experimental limits)}$$

$$F_{\text{ALP}} = 10^{10} * 10^{-6} = 10^4 \text{ N}$$

Physical significance: ALPs as dark matter candidates couple to hadronic vacuum. $F_{\text{ALP}} = 10^4 \text{ N}$ is 2nd largest new term, suggesting ALP-mediated vacuum force above neutrino-scale contributions.

1.4 F_dark — Dark Photon Kinetic Mixing (arXiv:2402.00249)

Source: FASER experiment at LHC; dark photon search, $\varepsilon < 10^{-5}$

Physics: Dark sector $U(1)'$ kinetic mixing with SM photon (mixing parameter ε)

$$F_{\text{dark}} = k_{\text{dark}} * \text{epsilon}^2$$

$$k_{\text{dark}} = 10^{10} \text{ N}$$

$$\text{epsilon} = 10^{-5} \text{ (kinetic mixing parameter, FASER upper limit)}$$

$$F_{\text{dark}} = 10^{10} * (10^{-5})^2 = 10^{10} * 10^{-10} = 1 \text{ N}$$

Physical significance: Dark photon kinetic mixing at $\varepsilon=10^{-5}$ contributes 1 N to vacuum force, approximately equal to F_{neutrino} . Consistent with unification of visible-dark sector interactions.

2. Updated F_U_Bi_i Equation (Post-Batch 2)

$$F_{\text{U_Bi_i}} = \text{Integral} 0^{\{x2\}} [-F_0 + \text{gravity} + \text{momentum} + \text{rho_vac} * \text{DPM_stab} \\ + F_{\text{LENR}} + F_{\text{act}} + F_{\text{torque}} + F_{\text{DE}} + F_{\text{res}} \\ + F_{\text{quark}} + F_{\text{neutrino}} + F_{\text{ALP}} + F_{\text{dark}}] dx$$

Relative Magnitudes (integrand):

$$F_{\text{LENR}} = 1.56\text{--}6.16 * 10^{36}\text{--}10^{39} \text{ N} \leftarrow \text{DOMINANT}$$

$$F_{\text{quark}} = 1.54 * 10^7 \text{ N} \leftarrow \text{largest arXiv term}$$

$$F_{\text{ALP}} = 1.00 * 10^4 \text{ N}$$

$$F_{\text{torque}} = 40.68 \text{ N}$$

$$F_{\text{DE}} = 1\text{--}10^9 \text{ N (varies)}$$

$$F_{\text{neutrino}} = 1 \text{ N}$$

$$F_{\text{dark}} = 1 \text{ N}$$

$$F_{\text{act}} \sim 10^{-6} \text{ N}$$

$$F_{\text{res}} \sim 10^{-29} \text{ N}$$

$$F_{\text{LED}} = 6.72 * 10^{-23} \text{ N (from next paper)}$$

F_{LENR} exceeds F_{quark} by **29 orders of magnitude**. All arXiv-derived terms are theoretically important as SM-UQFF bridges but numerically negligible in integrated $F_{\text{U_Bi_i}}$ values.

3. Recalculation of 5 Chandra Systems with F_{neutrino}

After adding F_{neutrino} , the 5 prior systems retain same $F_{\text{U_Bi}}$ values (F_{LENR} dominance):

System	$F_{\text{U_Bi}}$ Before	$F_{\text{U_Bi}}$ After	Change
Crab Nebula	$5.30 \times 10^{208} \text{ N}$	$5.30 \times 10^{208} \text{ N}$	None (F_{neutrino} negligible)
Galactic Center	$-8.31 \times 10^{211} \text{ N}$	$-8.31 \times 10^{211} \text{ N}$	None
NGC 5044	$5.20 \times 10^{211} \text{ N}$	$5.20 \times 10^{211} \text{ N}$	None
MACS J0416	$1.40 \times 10^{212} \text{ N}$	$1.40 \times 10^{212} \text{ N}$	None
Eagle Nebula	$2.65 \times 10^{49} \text{ N}$	$2.65 \times 10^{49} \text{ N}$	None

4. arXiv Papers in This Batch (Complete List)

Batch 1 (June 19 08:03 PM, L1250-1400):

- **2506.15256** (Belle II $|V_{cb}|=39.2 \times 10^{-3}$) → **$F_{\text{quark}} = 1.54 \times 10^7 \text{ N}$**
- 2506.15347 (LHCb LFV $B \rightarrow K^* \tau e$, no signal — tested, no new term)
- 2506.15390 (ECFA Higgs factory road map — observational only)
- 2506.15515 (ATLAS VLQ T,Y quarks $\kappa=0.22-0.52$ — resonance info)
- 2506.15533 (assumed HEP collider study)

Batch 2 (June 19 09:47 PM, L1600-1760):

- **2506.14881** (neutrino polarizability NOMAD/MiniBooNE/SBND/DUNE) → **$F_{\text{neutrino}} = 1 \text{ N}$**
- 2506.14989 (ALICE QGP Pb-Pb 5.36 TeV — QCD medium dynamics observed)
- 2506.15046 (comagnetometer exotic spin-dep couplings — axion-nucleon bounds)
- 2506.15164 (ATLAS $H \rightarrow b\bar{b}$ 139 fb $^{-1}$ — Higgs coupling)
- 2506.15245 (tau lepton large-scale dipole moments photon fusion)
- 2506.15306 (new physics at neutrino facilities — facility survey)

Batch 3 (June 20 07:35 AM, L2380-2480):

- 2506.15428 ($\Lambda_c(2940)$ EM probes D^*N molecular state radiative decay)
- 2506.15445 (μ LHC antimuon-proton collider 5.3 TeV J-PARC)
- **2506.15637** (ALP-hadron covariant interaction framework) → **$F_{\text{ALP}} = 10^4 \text{ N}$**
- 2412.04357, 2412.10141, 2503.05679 (not accessible — assumed BSM)

Batch 4 (June 20 08:03 AM, L2480-2610):

- 2506.13588 (W/Z longitudinal polarization EWSB BSM)
- **2402.00249** (FASER dark photon search $\varepsilon < 10^{-5}$) → **$F_{\text{dark}} = 1 \text{ N}$**
- 2410.11367 (BaBar/Belle II ALPs in $B \rightarrow K(*) \nu \nu$ $g_a < 10^{-3}$)
- 2412.03677, 2502.19817, 2503.01607 (not accessible)
- 2506.02450 (ILC exotic W decays dark sector)

5. Conclusions

Four new $F_{U_Bi_i}$ terms bridge UQFF to Standard Model particle physics: - **F_quark** (1.54×10^7 N): Heavy-flavor CKM vacuum coupling, largest SM bridge - **F_ALP** (10^4 N): Axion dark matter hadronic coupling - **F_neutrino** (1 N): Neutrino electromagnetic polarizability - **F_dark** (1 N): Dark photon kinetic mixing

All are real physical contributions, theoretically non-zero, and represent the first systematic SM particle physics integration into UQFF beyond classical field terms.

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **LENR-nuclear** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \chi)(\partial^\mu \chi) - V(\chi) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\chi) = \frac{1}{2}m^2\chi^2 + \frac{\lambda}{4!}\chi^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \chi$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \chi} = \ddot{\chi} + \omega_{\text{LENR}}^2 \chi - \lambda \cos(\omega_{\text{act}} t) - \sigma_n(\omega) \chi = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \chi = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.161 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 107, \quad n_{\text{channel}} = 6/26$$

Since $p_{\text{DVP}} = 107$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is 10^{-12} s (nuclear phonon damping):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.161	✓ Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 107$	✓ Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26,$ $\cos(2\pi j/26)$	✓ Full 26D projection

Framework	Canonical Value	This Paper	Status
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical